

STEP 1 — Install and Load spaCy Model

```
!pip install spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/py
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/pyth
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/pyt
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/pyth
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/l
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/pytho
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/py
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/pyt
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/pyt
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.1
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.1
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.1
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/pyt
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/p
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/pyt
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.1
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pyth
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-p
Collecting en-core-web-sm==3.8.0
```

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0.tar.gz 12.8/12.8 MB 68.1 MB/s eta 0:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
import spacy

nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")

spaCy model loaded successfully
```

```
sentences = [ "The cat chased the mouse.", "She is reading a book.", "The weather is
very pleasant today.", "The engineer designed a new system.", "The teacher explained
the lesson clearly.", "John and Mary went to the market.", "I studied hard but the exam
was difficult.", "Although it was raining, they played football.", "If you work hard, you
will succeed.", "Why are you late today?", "Did she complete the project?", "The report
was written by the manager.", "The software was developed by the company.",
"Artificial intelligence is transforming technology.", "Students are learning natural
language processing." ]
```

```
sentences = [
"The cat chased the mouse.",
"She is reading a book.",
"The weather is very pleasant today.",
"The engineer designed a new system.",
"The teacher explained the lesson clearly.",
"John and Mary went to the market.",
"I studied hard but the exam was difficult.",
"Although it was raining, they played football.",
"If you work hard, you will succeed.",
"Why are you late today?",
"Did she complete the project?",
"The report was written by the manager.",
"The software was developed by the company.",
"Artificial intelligence is transforming technology.",
"Students are learning natural language processing."
]
```

STEP 3 — Process Sentences using spaCy

```
docs = [nlp(sentence) for sentence in sentences]

for doc in docs:
    print("Sentence:", doc.text)
    print("Tokens:", [token.text for token in doc])
    print()
```

```
Sentence: The cat chased the mouse.
Tokens: ['The', 'cat', 'chased', 'the', 'mouse', '.']

Sentence: She is reading a book.
Tokens: ['She', 'is', 'reading', 'a', 'book', '.']
```

Sentence: The weather is very pleasant today.

Tokens: ['The', 'weather', 'is', 'very', 'pleasant', 'today', '.']

Sentence: The engineer designed a new system.

Tokens: ['The', 'engineer', 'designed', 'a', 'new', 'system', '.']

Sentence: The teacher explained the lesson clearly.

Tokens: ['The', 'teacher', 'explained', 'the', 'lesson', 'clearly', '.']

Sentence: John and Mary went to the market.

Tokens: ['John', 'and', 'Mary', 'went', 'to', 'the', 'market', '.']

Sentence: I studied hard but the exam was difficult.

Tokens: ['I', 'studied', 'hard', 'but', 'the', 'exam', 'was', 'difficult', '.']

Sentence: Although it was raining, they played football.

Tokens: ['Although', 'it', 'was', 'raining', ',', 'they', 'played', 'football']

Sentence: If you work hard, you will succeed.

Tokens: ['If', 'you', 'work', 'hard', ',', 'you', 'will', 'succeed', '.']

Sentence: Why are you late today?

Tokens: ['Why', 'are', 'you', 'late', 'today', '?']

Sentence: Did she complete the project?

Tokens: ['Did', 'she', 'complete', 'the', 'project', '?']

Sentence: The report was written by the manager.

Tokens: ['The', 'report', 'was', 'written', 'by', 'the', 'manager', '.']

Sentence: The software was developed by the company.

Tokens: ['The', 'software', 'was', 'developed', 'by', 'the', 'company', '.']

Sentence: Artificial intelligence is transforming technology.

Tokens: ['Artificial', 'intelligence', 'is', 'transforming', 'technology', '.']

Sentence: Students are learning natural language processing.

Tokens: ['Students', 'are', 'learning', 'natural', 'language', 'processing', '.']

STEP 4 — Extract Dependency Relations

```
for doc in docs:
    print("Sentence:", doc.text)
    print("Token\tLemma\tPOS\tDependency\tHead")

    for token in doc:
        print(token.text, token.lemma_, token.pos_, token.dep_, token.head)

    print("\n")
```

```

Did do AUX aux complete
she she PRON nsubj complete
complete complete VERB ROOT complete
the the DET det project
project project NOUN dobj complete
? ? PUNCT punct complete

```

Sentence: The report was written by the manager.

```

Token   Lemma   POS      Dependency   Head
The the DET det report
report report NOUN nsubjpass written
was be AUX auxpass written
written write VERB ROOT written
by by ADP agent written
the the DET det manager
manager manager NOUN pobj by
. . PUNCT punct written

```

Sentence: The software was developed by the company.

```

Token   Lemma   POS      Dependency   Head
The the DET det software
software software NOUN nsubjpass developed
was be AUX auxpass developed
developed develop VERB ROOT developed
by by ADP agent developed
the the DET det company
company company NOUN pobj by
. . PUNCT punct developed

```

Sentence: Artificial intelligence is transforming technology.

```

Token   Lemma   POS      Dependency   Head
Artificial artificial ADJ amod intelligence
intelligence intelligence NOUN nsubj transforming
is be AUX aux transforming
transforming transform VERB ROOT transforming
technology technology NOUN dobj transforming
. . PUNCT punct transforming

```

Sentence: Students are learning natural language processing.

```

Token   Lemma   POS      Dependency   Head
Students student NOUN nsubj learning
are be AUX aux learning
learning learn VERB ROOT learning
natural natural ADJ amod language
language language NOUN compound processing
processing processing NOUN dobj learning
. . PUNCT punct learning

```

STEP 5 — Identify Sentence Structure

```

for doc in docs:
    print("Sentence:", doc.text)

```

```
subject = [token.text for token in doc if token.dep_ == "nsubj"]
verb = [token.text for token in doc if token.dep_ == "ROOT"]
obj = [token.text for token in doc if token.dep_ == "dobj"]

print("Subject:", subject)
print("Verb:", verb)
print("Object:", obj)
print()
```

```
Verb: ['designed']
Object: ['system']
```

```
Sentence: The teacher explained the lesson clearly.
Subject: ['teacher']
Verb: ['explained']
Object: ['lesson']
```

```
Sentence: John and Mary went to the market.
Subject: ['John']
Verb: ['went']
Object: []
```

```
Sentence: I studied hard but the exam was difficult.
Subject: ['I', 'exam']
Verb: ['studied']
Object: []
```

```
Sentence: Although it was raining, they played football.
Subject: ['it', 'they']
Verb: ['played']
Object: ['football']
```

```
Sentence: If you work hard, you will succeed.
Subject: ['you', 'you']
Verb: ['succeed']
Object: []
```

```
Sentence: Why are you late today?
Subject: ['you']
Verb: ['are']
Object: []
```

```
Subject: [ 'intelligence' ]  
Verb: [ 'transforming' ]  
Object: [ 'technology' ]
```

```
Sentence: Students are learning natural language processing.  
Subject: [ 'Students' ]  
Verb: [ 'learning' ]  
Object: [ 'processing' ]
```

STEP 6 — Dependency Tree Visualization

```
from spacy import displacy  
  
for doc in docs[:5]:  
    displacy.render(doc, style="dep", jupyter=True)
```

